Heart Rate Responses During Small-Sided Games and Short Intermittent Running Training in Elite Soccer Players: A Comparative Study

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Abstract

Dellal, A, Chamari, K, Pintus, A, Girard, O, Cotte, T, and Keller, D. Heart rate responses during small-sided games and short intermittent running training in elite soccer players: a comparative study. *J Strength Cond Res* 22(5): 1449–1457, 2008—The purpose of this study was to compare heart rate (HR) responses within and between physical controlled (short-duration intermittent running) and physical integrated (sided games) training methods in elite soccer players. Ten adult male elite soccer players (age, 26 ± 2.9 years; body mass, 78.3 ± 4.4 kg; maximum HR [HRmax], 195.4 ± 4.9 b·min^{-1} and velocity at maximal aerobic speed (MAS), 17.1 ± 0.8 km·h^{-1}) performed different short-duration intermittent runs, e.g., 30–30 (30 seconds of exercise interspersed with 30 seconds of recovery) with active recovery, and 30–30, 15–15, 10–10, and 5–20 seconds with passive recovery, and different sided games (1 versus 1, 2 versus 2, 4 versus 4, 8 versus 8 with and without a goalkeeper, and 10 versus 10). In both training methods, HR was measured and expressed as a mean percentage of HR reserve (%HRres). The %HRres in the 30–30-second intermittent run at 100% MAS with active recovery (at 9 km·h^{-1} with corresponding distance) was significantly higher than that with passive recovery (85.7% versus 77.2% HRres, respectively, $p < 0.001$) but also higher than the 1 versus 1 ($p < 0.01$), 4 versus 4 ($p \leq 0.05$), 8 versus 8 ($p < 0.001$), and 10 versus 10 ($p < 0.01$) small-sided games. The %HRres was 2-fold less homogeneous during the different small-sided games than during the short-duration intermittent running (intersubjects coefficient of variation [CV] = 11.8% versus 5.9%, respectively). During the 8 versus 8 sided game, the presence of goalkeepers induced an ~11% increase in %HRres and reduced homogeneity when compared to games without goalkeepers (intersubject CV = 15.6% versus 8.8%). In conclusion, these findings showed that some small-sided games allow the HR to increase to the same level as that in short-duration intermittent running. The sided game method can be used to bring more variety during training, mixing physical, technical, and tactical training approaching the intensity of short-duration intermittent running but with higher intersubject variability.

Key Words interval training, reduced games, physical integrated training, performance

Introduction

Endurance training effects of short-duration intermittent runs have been reported in athletes (3,9,23) with improved maximal oxygen consumption ($\dot{V}O_2max$) and delayed fatigue compared to continuous running methods (10,15). Balsom et al. (5,8) have shown that short-duration intermittent training allows limited lactate production and increased creatine phosphate metabolism during intermittent exercise. Creatine phosphate (5,21,32,37) and muscle glycogen (8,12,34) were described as the most important energy substances for this type of training. The effects of training with short-duration intermittent running in elite soccer players have been analyzed for years (6,11). These studies have shown that this type of training method has the potential to improve the players' endurance. A high level of aerobic capacity in elite soccer allows improved field performance, i.e., greater

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participation in the game through greater covered distance, involvement with the ball, and playing intensity (2,25,41).

Kirkendall (29) described soccer as the amount of game phases at 4 versus 4 or less, on reduced areas. Other authors have shown that during specific aerobic training, it is possible to reach heart rates (HRs) similar to running interval training (26,27). One of the main differences between these 2 training methods is that the presence of the ball during small-sided games allows the concomitant improvement of technical and tactical skills with greater motivation of the players (20). Nevertheless, the players are relatively free during the sided games and their effort is highly dependent on their level of individual motivation. It is also suggested that this level of motivation is generally increased in soccer when goalkeepers are included, inducing the players to play to score (1). In this context, recent studies have reported conflicting results concerning the physiological impact of different small-sided games. Comparing the activity of soccer players during two reduced games (5 versus 5 and 11 versus 11), Allen et al. (1) noted that despite an equal distance jogged, changes in HR during the activity and the number of ball contacts during the 5 versus 5 game was significantly greater than those during the 11 versus 11 game. The number of players, player activity, and the presence of goalkeepers have an impact on HR responses (1). During sided games, coaches cannot accurately control the activity of their players, and it is not very clear to what extent this training modality has the potential to produce the same physiological responses as does short-duration intermittent running training. In this context, it was reported that HR monitoring during various sided games was considered valid to reflect the intensity of the soccer players' activity (26).

Therefore, the purpose of the present study was to compare HR responses within and between generic physical (short-duration intermittent running) and integrated physical (sided games) training in elite soccer players. First, it is hypothesized that HR responses are similar during some sided games and short-duration intermittent runs. Then, it is hypothesized that the intersubject homogeneity of HR responses during sided games is less than during intermittent running and that the presence of goalkeepers increases game intensity.

METHODS

Experimental Approach to the Problem

During the experimental period, 2 study-related sessions were implemented per week: 1 short-duration intermittent running and 1 sided game session. Sessions and exercises were scheduled as shown in Table 1 (as commonly used in high-level soccer players). The experiment was performed just after mid-season (7 days of rest), i.e., during the sixth and the seventh months after the beginning of the season. Each week the medical staff checked the players' hormone levels, body mass, and HR activity (variability in resting HR). The players were healthy and did not have any recent injury. Subjects

TABLE 1. Experimental set-up.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
		Week 1				
	VAMEVAL test		Rest		Official game	
		Week 2				
	2 vs. 2	10–10-s at 110% $\dot{V}O_2max$	Rest			Official game
		Week 3				
	30–30-s at 100% $\dot{V}O_2max$ AR	10 vs. 10 GK	Rest		Official game	
		Week 4				
		30–30-s at 100% $\dot{V}O_2max$ PR	Rest	8 vs. 8 GK	Official game	
		Week 5				
		1 vs. 1	Rest	5–20-s at 120% $\dot{V}O_2max$	Official game	
		Week 6				
	8 vs. 8		Rest			
		Week 7				
15–15-s at 100% $\dot{V}O_2max$		4 vs. 4 GK			Official game	

VAMEVAL; $\dot{V}O_2max$ = lowest velocity associated with maximum oxygen consumption or maximum aerobic speed; *x* versus *y* = sided games with *x* against *y* players; GK = presence of goalkeepers; PR = passive recovery; AR = active recovery (the soccer players had to jog at 9 km·h^{-1}).

were first tested with a maximal field test in order to measure maximal aerobic speed or the velocity at maximum oxygen consumption ($\dot{V}o_2max$) and maximum HR (HRmax). Each player was verbally encouraged to maximally perform during the overall protocol during which players continued to train normally, i.e., 5–7 training sessions and 1 competition match per week (Table 1). The maximal field test and the most difficult experimental training sessions took place on Tuesday or Wednesday in order to be far enough away from the official games on Sunday. Generally the players played an official game on Saturday or Sunday. On Monday, the training consisted of recovery sessions consisting of jogging at 12 $km \cdot h^{-1}$ for 20 minutes and therapeutic actions (e.g., massage, sauna, thermoaction). On Tuesday, 2 training sessions were performed: muscle strengthening in the morning (e.g., general muscle work) and a technical-tactical (e.g., technical circuit) training session in the afternoon. On Wednesday, there was a mixed session with an aerobic power session (interval training and sided games) and a technical-tactical workout. On Thursday, no training was performed. On Friday, 2 technical-tactical training sessions with a reactivity session (e.g., multiple short-burst sprints) were performed. An appropriate standardized warm-up (e.g., general physical preparation with articular and muscular mobilization) was performed before each training session.

Subjects

Ten male elite soccer players belonging to a French first league senior team volunteered to participate in the study. Players' physical characteristics are presented in Table 2. The protocol was approved by the local university ethics committee, and all subjects gave written informed consent before participating. The subjects could withdraw from the study at any time and were informed about the protocol details without being informed about the aim of the study.

Procedures

Field Testing: The VAMEVAL Test. The VAMEVAL field test (14) was performed in the same afternoon for all subjects from 5:00 to 6:30 PM in ambient conditions of 16°C, 1019 mm Hg atmospheric pressure, and 46% relative humidity. The test was performed on a natural grass soccer field, and subjects wore soccer uniforms. Subjects were familiar with this testing procedure as it was often used to set the training pace during training sessions.

The VAMEVAL maximal incremental running test is very similar to the University of Montreal Field Test (14). It begins with a running speed of 8 $km \cdot h^{-1}$ with consecutive speed increases of 0.5 $km \cdot h^{-1}$ each minute until exhaustion. The subjects adjusted their running velocity to auditory signals at 20-m intervals, delineated by visual marks along a 200-m long track. This test estimated the subjects' maximal aerobic speed (MAS) and measured the HRmax. The MAS is the velocity in $km \cdot h^{-1}$ of the last 1-minute stage completed by the subject. The uncompleted 1-minute stages were not taken into account (30).

Short-Duration Intermittent Running. All subjects completed all the short duration intermittent running sessions presented in Table 3. During these sessions, running distances were individualized based on the subject's measured MAS. In case of active recovery, players had to jog at 9 $km \cdot h^{-1}$ (with corresponding distance). Each exercise protocol was performed in different weeks and on a natural grass field.

Small-Sided Games. All subjects performed all the different sided games presented in Table 4. The different sided games were performed with or without a goalkeeper in smaller areas on a natural grass field. With goalkeepers, the sided games consisted of playing a match, whereas without goalkeepers, the sided games consisted of keeping the ball despite the opponents' efforts to take the ball. Several balls were placed all around the area of the sided games for immediate availability in order to avoid stopping the game. In both modes of training, players were allowed hydration ad libitum.

HR Measurements and Calculations. The athletes' HR (recorded in 5-second intervals) was continuously recorded using HR monitors (Polar S-810; Polar-Electro, Kempele, Finland) during each training modality. The HR time course was

TABLE 2. Characteristics of the subjects (mean ± *SD*).

Subjects (N = 10)	Age (y)	Body mass (kg)	Body height (cm)	MAS ($km \cdot h^{-1}$)	HRmax, ($b \cdot min^{-1}$)	HRrest ($b \cdot min^{-1}$)	HRres ($b \cdot min^{-1}$)
Values	26.0 ± 2.9	78.3 ± 4.4	181.4 ± 5.9	17.1 ± 0.8	195.4 ± 4.9	52.0 ± 3.8	144.3 ± 5.6

MAS = maximal aerobic speed; HRmax = maximum heart rate observed at the end of the VAMEVAL test; HRrest = minimum HR observed when the athletes laid on a bed for 10 minutes at 10:30 AM; HRres = heart rate reserve, i.e., difference between the maximum and resting heart rate.

Table 3. Short-duration intermittent running characteristics.

Intermittent training (working time–recovery time) (s)	Work intensity	No. × duration of series	Intraseries recovery	Interseries recovery (min)	Total duration of the session (min)
30–30 PR	100% $\dot{V}O_2max$	2 × 10 min	Passive	10 (passive)	30
30–30 AR	100% $\dot{V}O_2max$	2 × 10 min	Active	10 (passive)	30
15–15	100% $\dot{V}O_2max$	2 × 10 min	Passive	8 (passive)	28
10–10	110% $\dot{V}O_2max$	2 × 7 min	Passive	6 (passive)	20
5–20	120% $\dot{V}O_2max$	1 × 7 min, 5 s	Passive	–	7

PR = passive recovery; $\dot{V}O_2max$ = lowest velocity associated with maximum oxygen consumption or maximum aerobic speed; AR = active recovery (the soccer players had to jog at 9 $km \cdot h^{-1}$).

analyzed from the beginning to the end of each short-duration intermittent running session and each sided game. Concerning the resting HR, the athletes laid on a bed for 10 minutes at 10:30 AM. The resting HR value corresponded to the minimal HR observed during this 10-minute period. During the VAMEVAL field test, the highest averaged value of 3 consecutively recorded HRs (15 seconds) was considered the HRmax. The percentage of HR reserve (%HRres) was calculated by the following formula (28): %HRres = (exercise mean HR − resting HR)/(HRmax − resting HR) × 100.

For each short-duration intermittent running session and each sided game, the %HRres and the intersubject coefficient of variation (CV) were calculated.

Statistical Analyses

Values are expressed as mean ± *SD*. The normality distribution of the data was checked with the Kolmogorov-Smirnov test. After confirming normal distribution, a 1-way repeated-measures analysis of variance was used to evaluate the differences in %HRres within and between sided games and short-duration intermittent running training modalities. A p value ≤ 0.05 was considered as statistically significant.

Results

Within Intermittent Exercise and Within Sided Game Comparison

Compared to the 30–30-s intermittent exercise at 100% $\dot{V}O_2max$ with passive recovery (PR), a significant 9.1% increase ($p \leq 0.05$) of the %HRres was noted with respect to the same exercise with active recovery (AR).

There was no significant difference between the 30–30-s and 15–15-s intermittent exercise performed with the same characteristics of interseries type of recovery (i.e., passive), work intensity, series duration, and total duration of the session.

With the same characteristics, the intensity of the 8 versus 8 significantly increased with the presence of goalkeepers (+10.7%HRres). However, the game intensity of the soccer players during the 8 versus 8 sided game with goalkeepers was less homogeneous than without goalkeepers (Table 5).

Intermittent Exercise and Sided Game Comparison

The overall soccer players' HR response was 2-fold less homogeneous during sided games (intersubject CV = 11.8%)

Table 4. Characteristics of sided games.

Sided game	No. of goalkeepers	Game area (m)	No. × duration of series	Interseries recovery	Total duration of the session
1 vs. 1	0	10 × 10	4 × 1 min, 30 s	1 min, 30 s (passive)	10 min, 30 s
2 vs. 2	0	20 × 20	6 × 2 min, 30 s	2 min, 30 s (passive)	27 min, 30 s
4 vs. 4 GK	2	30 × 25	2 × 4 min	3 min (passive)	11 min
8 vs. 8 GK	2	60 × 45	2 × 10 min	5 min (passive)	25 min
8 vs. 8	0	60 × 45	4 × 4 min	3 min (passive)	25 min
10 vs. 10 GK	2	90 × 45	3 × 20 min	5 min (passive)	70 min

Total duration of the session = the end of this duration corresponded to the end of the last series of the reduced game; GK = presence of goalkeepers.

Table 5. Percentage of heart rate reserve (%HRres) during the different sided games and during the different short-duration intermittent running sessions.

	Training method										
	Sided games						Intermittent				
	1 vs. 1	2 vs. 2	4 vs. 4 + GK	8 vs. 8 + GK	8 vs. 8	10 vs. 10 + GK	10–10 110% $\dot{V}O_2max$ PR	30–30 100% $\dot{V}O_2max$ PR	30–30 100% $\dot{V}O_2max$ AR	15–15 100% $\dot{V}O_2max$ PR	5–20 120% $\dot{V}O_2max$ PR
%HRres	77.6 ± 8.6	80.1 ± 8.7	77.1 ± 10.7	80.3 ± 12.5	71.7 ± 6.3	75.7 ± 7.9	85.8 ± 3.9	77.2 ± 4.6	85.7 ± 4.5	76.8 ± 4	80.2 ± 6.8
Intersubject CV (%)	11.12	10.83	13.87	15.60	8.79	10.40	4.50	5.97	5.27	5.20	8.50

GK = presence of goalkeepers; $\dot{V}O_2max$ = lowest velocity associated with maximum oxygen consumption; PR = passive recovery; AR = active recovery (the soccer players had to jog at 9 $km \cdot h^{-1}$); %HRres = percentage of heart rate reserve; CV = coefficient of variation.

compared to intermittent running training methods (intersubject CV = 5.9%).

The HR response was significantly greater during the 30–30-s intermittent exercise with AR than during the 1 versus 1 ($p < 0.01$), 4 versus 4 ($p \leq 0.05$), 8 versus 8 ($p < 0.001$), and 10 versus 10 ($p < 0.01$) sided games, whereas with PR, the HR response was not significantly different during all the sided games (Figure 1). Concerning the 10–10-s intermittent exercise with PR, the HR response was also significantly greater than during the 1 versus 1 ($p \leq 0.05$), the 8 versus 8 ($p < 0.01$), and the 10 versus 10 ($p \leq 0.05$) sided games (Figure 1), whereas during the 2 versus 2 and the 8 versus 8 sided games with goalkeepers, the HR response was not significantly different during all the different short-duration intermittent running sessions (Figure 2).

Discussion

The purpose of this study was to compare HR responses within and between controlled generic (short-duration intermittent running) and integrated (sided games) training methods in elite soccer players.

Although some authors (6,16,25) demonstrated the effectiveness of short-duration intermittent running, others have suggested sided games as soccer-specific aerobic endurance training (4,7,26,27). This study shows that the 2 versus 2 and 8 versus 8 sided games with goalkeepers induced similar HR responses than did several intermittent exercise sessions, i.e., 5–20-s at 120% $\dot{V}O_2max$, 10–10-s at 110% $\dot{V}O_2max$, 15–15-s at 110% $\dot{V}O_2max$, and 30–30-s at 100% $\dot{V}O_2max$ with AR or PR. Thus, one may argue that these 2 sided games can be used for soccer-specific aerobic endurance training with the advantages of multifactorial training.

The other sided games, i.e., the 1 versus 1, 4 versus 4, and 10 versus 10 induced the same HR responses as the 15–15-s at 110% $\dot{V}O_2max$ and 30–30-s at 100% $\dot{V}O_2max$ with PR intermittent runs. These results suggest that all sided games cannot allow practicing specific aerobic endurance training with a minimum of required exercise intensity. Concerning the 10–10-s intermittent exercise at 110% $\dot{V}O_2max$ with PR, HR was also significantly higher than during the 1 versus 1. Balsom (4) found similar results with different 3 versus 3 sided games on an area of 33 × 22 m in elite soccer players. He reported that the continuous exercise without a ball, the intermittent exercise (30–30-s intermittent exercise at 100% $\dot{V}O_2max$), and the 3 versus 3 sided game imposed similar cardiovascular stress. Thus, Balsom (4) concluded that these sided games could improve soccer players' endurance. All these results confirm the recommendation made to coaches to use this training method as a specific physical training, also called integrated training. Indeed, sided games produce similar cardiovascular stress as other intermittent exercises specifically designed to improve athletes' endurance. One of the differences between sided games and short-duration intermittent running training methods is the presence of the

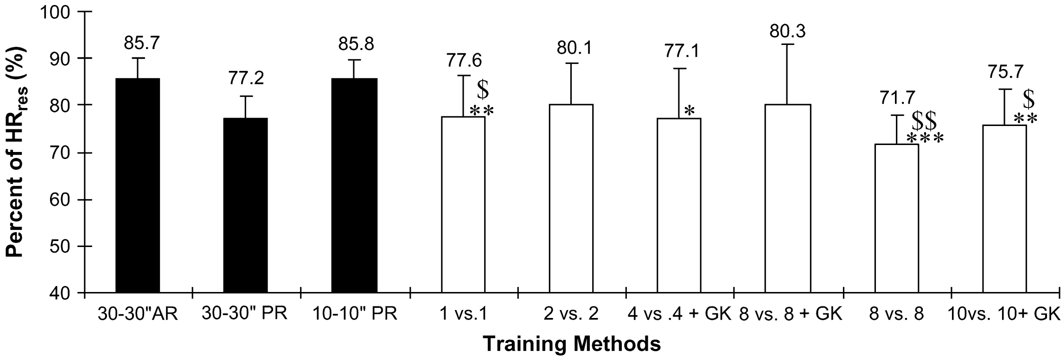

Figure 1. Comparison of the percentage of heart rate reserve (%HRres) during the 30–30-s intermittent running at 100% of the lowest velocity at the maximum oxygen consumption ($\dot{V}O_2max$) with active recovery (AR) (the soccer players had to jog at 9 km·h^{-1}) and passive recovery (PR), the 10–10-s intermittent running at 110% $\dot{V}O_2max$, and the different sided games. *Significant difference between the 30–30-s intermittent exercise with AR and the sided games. *$p \leq 0.05$; **$p < 0.01$; ***$p < 0.001$. \$Significant difference between the 10–10-s intermittent exercise with PR and the sided games. \$$p \leq 0.05$; \$\$$p < 0.01$; GK = presence of goalkeepers; *x* versus *y* = sided games with *x* against *y* players.

ball, which imposes a specific activity and allows the concomitant improvement of technical and tactical skills with high player motivation.

Nevertheless, the soccer players' HR response was less homogeneous during the different sided games compared to intermittent running (intersubject CV = 11.8% versus 5.9%, respectively). Indeed, the activity of the soccer players was not totally controlled by the staff because the moves of soccer players were different depending on their experience, their position during the competition game, the movements of the opponents, and/or their motivation (38,39).

Coaches may use sided game training, which requires technical, tactical, and physical aspects at the same time, making it very similar to the soccer game itself. Therefore, the physical effect was variable according to the game. A sided game change in characteristics implies a change in physiological impact. The game area, the number of players, game instructions, the number and duration of the series, the total duration of the session, and the presence of goalkeepers directly influence the activity of the players and the physiological impact (4). Allen et al. (1) compared the activity of soccer players during 2 sided games (5 versus 5 and 11 versus 11). Despite an equal distance jogged, these authors showed that the players' activity during the 5 versus 5 was significantly greater than that during the 11 versus 11. In addition, they found that the number of ball contacts was also significantly higher with the smallest number of players, i.e., 5 versus 5. The number of players is important for physically

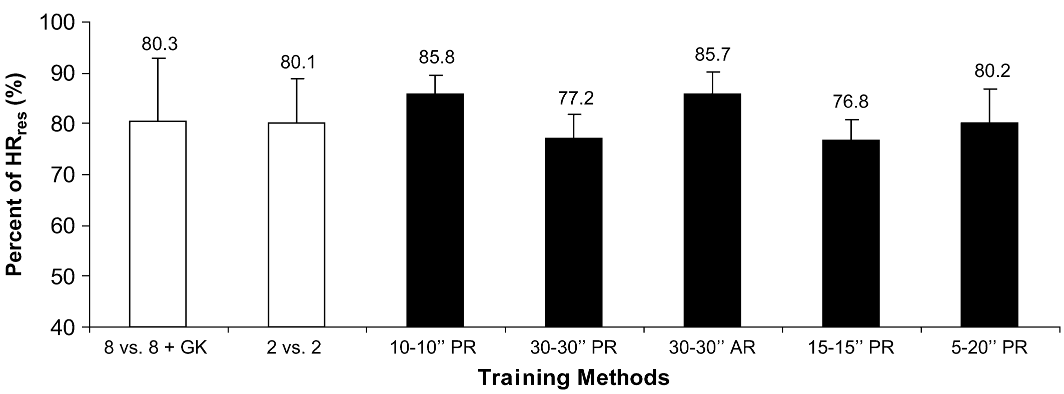

Figure 2. Comparison of the percentage of heart rate reserve (%HRres) during the 8 versus 8 with goalkeepers sided game, the 2 versus 2 sided game, and the different short-duration intermittent running. GK = presence of goalkeepers; *x* versus *y* = sided games with *x* against *y* players; $\dot{V}O_2max$ = the lowest velocity at the maximum oxygen consumption or maximum aerobic speed.

specific integrated training. Indeed, Rampinini et al. (35) showed that the intensity of sided games increases while the number of players decreases. Nevertheless, this is also dependent on the playing area, with the game intensity decreasing when the available game area is decreased (35).

Another key factor in influencing the sided games is the presence of the goalkeepers, which increases the HR of the soccer players during sided games. In the present study, the 8 versus 8 sided games with goalkeepers showed a 12% increase in the %HRres compared to the same game without goalkeepers. The aims of scoring and protecting their own goalkeepers may have imposed greater activity on the soccer players probably because of higher motivation (38,39). The activity of the soccer player was less homogeneous during 8 versus 8 sided game with goalkeepers than without goalkeepers (intersubject CV = 15.6% versus 8.8%, respectively). It is possible that some players are more motivated than others by the presence of goalkeepers, explaining such a difference.

The physiological impacts of sided games were less homogeneous than those of intermittent running. This difference was increased with the presence of goalkeepers, the modification of the number of players, and the decrease in area. The sided games characteristics and rules have to be chosen with care because each element can influence the activity of the soccer players (35).

Thus, if the staff aims to totally control the physical training effects for the majority of the players involved and thus reduce interplayer variability, they have to use intermittent running sessions. Some authors have reported that high-intensity intermittent exercise can both improve the $\dot{V}O_2max$ and the performance of the soccer players (7,28). In this study, the most intensive intermittent exercise was the 10–10-s (110% of $\dot{V}O_2max$) and the least intensive was the 15–15-s intermittent exercise (100% of $\dot{V}O_2max$). This type of short-duration intermittent exercise solicits the anaerobic and aerobic mechanisms at the same time (23,40). In addition, Dupont and Berthoin (17) have shown that the 15–15-s intermittent exercise at 120% $\dot{V}O_2max$ was the best intermittent exercise to improve $\dot{V}O_2max$. The present study showed that the 15–15-s intermittent exercise at a lower intensity, i.e., 100% $\dot{V}O_2max$, was insufficient to induce a high HR response, showing the importance of the choice of exercise intensity in intermittent training. These results confirm the fact that the development of $\dot{V}O_2max$ requires an exercise intensity higher than 100% of $\dot{V}O_2max$ when short-duration intermittent running is considered (22,31,36).

Millet et al. (33) showed that a 30–30-s (at 100% $\dot{V}O_2max$ for 10 minutes) intermittent exercise with PR allows 54 seconds more than 95% of $\dot{V}O_2max$) and ~150-s more than 90% of $\dot{V}O_2max$. A 30–30-s intermittent exercise at 100% $\dot{V}O_2max$ with AR is probably a training design that could produce improvement in $\dot{V}O_2max$ in soccer players (16). It could solicit the anaerobic and aerobic mechanisms at the same time (33). However, according to the data from the present study, for the same intermittent running training modality—30–30-s intermittent exercise at 100% $\dot{V}O_2max$, the difference in %HRres was about 9.1% between the PR and AR. This indicates that the type of recovery greatly determines the physiological responses of an athlete during a 30–30-s intermittent exercise at 100% $\dot{V}O_2max$. Dupont and Berthoin (17) have shown that the AR could be less efficient than the PR to optimize $\dot{V}O_2max$, because of the fact that AR enhances blood lactate removal in comparison with PR (13,24,33). Moreover, PR could allow greater reoxygenation of myoglobin and hemoglobin than AR (18,19). The results of this latter study showed that there were no significant differences between AR and PR concerning blood lactate concentration (12.6 $\pm$ 1 versus 13.1 $\pm$ 2.7 $mmol \cdot L^{-1}$), peak HR (183.3 $\pm$ 13.9 versus 182.5 $\pm$ 15 $b \cdot min^{-1}$) and average HR (66.6 $\pm$ 11.9; 165.6 $\pm$ 14.6 $b \cdot min^{-1}$, respectively). However, these authors have shown that there was a significant difference between AR and PR with regard to overall metabolic power.

Some of the limitations of the present study are the absence of randomization and the stability of the day-to-day HR response. Because the subjects were elite first division players, it was not possible to randomize the order of the sided games and intermittent runs performed. Randomization would have avoided any order effect of any exercise on any other one. Nevertheless, randomizing is much easy to implement in some studies and almost impossible in others. The proposal of randomization was rejected by the team coach as this would have imposed too much organization concerns and would have disturbed the regular training schedule of the staff. Concerning the potential bias of an eventual day-to-day HR variability, players were regularly checked for their resting HR variability and were coming back from a week of intraseason recovery. This does not impede the day-to-day HR variability bias, but might limit it. In this context, several other studies were conducted on soccer player training using HR as a physiological variable to control exercise intensity. Despite that, HR variability has to be taken into consideration when interpreting the results of the present study.

To summarize, our findings support the idea that some small-sided games enable HR responses to increase, similarly to some short-duration intermittent runs shown to be effective in enhancing player endurance. Nevertheless, some other sided games cannot increase the HR responses at the same level as the described intermittent exercises methods. Sided games show greater interplayer HR variability, and the presence of goalkeepers during sided games increases the physiological impact. Further studies are needed to compare the effects of these 2 training methods on the enhancement of soccer player endurance.

Practical Applications

Previous studies demonstrated the fact that intermittent exercise can be used to improve $\dot{V}O_2max$, but few studies have compared these training methods with specific ball training like sided games. The results of this study show that it is possible to use some sided games for physically integrated training approaching the intensity of the player's activity during short-duration intermittent running. However, the main difficulty during the sided games is to control the activity of the players. The choice of the number of players, the presence of goalkeepers, the playing area, and game instructions directly affect HR responses. According to the training objectives, the choice is between controlled physical training (short-duration intermittent running) and physically integrated training (sided games) in which the group activity varies more.

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References

1. Allen, JD, Butterfly, R, Welsh, MA, and Wood, R. The physical and physiological value of 5-a-side soccer training to 11-a-side match play. *J Hum Movement Stud* 34: 1–11, 1998.
2. Apor, P. Successful formulae for fitness training. In:*Science and Football.* Reilly, T, Lees, A, Davids, K, and Murphy, WJ, eds. London: E. & F.N. Spon, 1988. pp. 95–107.
3. Astrand, I, Astrand, PO, Christensen, EH, and Hedman, R. Intermittent muscular work. *Acta Physiol Scand* 48: 448–453, 1960.
4. Balsom, PD. *Precision Football.* Kempele, Finland: Polar Electro Oy, 1999.
5. Balsom, PD, Ekblom, B, Söderlund, K, Sjödin, B, and Hultman, E. Creatine supplementation and dynamic high-intensity intermittent exercise. *Scand J Med Sci Sports* 3: 143–149, 1993.
6. Bangsbo, J. The physiology of soccer— with special reference to intense intermittent exercise. *Acta Physiol Scand* 15(Suppl. 619): 1–156, 1994.
7. Bangsbo, J. Optimal preparation for the World Cup in soccer. *Clin Sports Med* 17: 697–709, 1998.
8. Bangsbo, J and Mizuno, M. Morphological and metabolic alterations in soccer players with detraining and retraining and their relation to performance. In: *Science and Football.* Reilly, T, Lees, A, Davids, K. and Murphy, WJ, eds. London: E. & F.N. Spon, 1988. pp. 114–124.
9. Billat, V, Flechet, B, Petit, B, Muriaux, G, and Koralsztein, JP. Interval training at VO_2max: effects on aerobic performance and overtraining markers. *Med Sci Sports Exerc* 31: 156–163, 1999.
10. Billat, V, Petit, B, and Koralsztein, JP. Calibration of the duration of the repetitions of a meeting of interval training at the speed associated with VO_2max in reference to the continuous time limits: effect on the physiological answers and the distance covered. *Sci Mot* 28: 13–20, 1996.
11. Bisciotti, NG, Sagnol, JM, and Filaire, E. Aspetti bioenergeticci della corsa frazionata [Bioenergetic aspects of interval training in soccer]. *Scuola Sport* 50: 21–27, 2000.
12. Bishop, NC, Gleeson, M, Nicholas, CW, and Ali, A. Influence of carbohydrate supplementation on plasma cytokine and neutrophil degranulation responses to high intensity intermittent exercise. *Int J Sport Nutr Exerc Metab* 12: 145–156, 2002.
13. Bonen, A and Belcastro, AN. Comparison of self-selected recovery methods on lactic-acid removal rates. *Med Sci Sports Exerc* 8: 176–178, 1976.
14. Cazorla, G and Léger, L. *How to Evaluate and Develop Your Aerobic Capacities. Tests of Race Shuttle and Test VAMEVAL.* Cestas, France: AREAPS, 1993. p. 123.
15. Christensen, EH, Hedman, R, and Saltin, B. Intermittent and continuous running. *Acta Physiol Scand* 50: 269–286, 1960.
16. Dupont, G, Akakpo, K, and Berthoin, S. The effect of in-season, high-intensity interval training in soccer players. *J Strength Cond Res* 18: 584–589, 2004.
17. Dupont, G and Berthoin, S. Time spent at a high percentage of VO_2max for short intermittent runs: active recovery versus passive recovery. *Can J Appl Physiol* 29:(Suppl.): S3–S16, 2004.
18. Dupont, G, Blondel, N, and Berthoin, S. Performance for short intermittent runs: active versus passive recovery. *Eur J Appl Physiol* 89: 548–554, 2003.
19. Dupont, G, Moalla, W, Guinhouya, C, Ahmaidi, S, and Berthoin, S. Passive versus active recovery during high intensity intermittent exercises. *Med Sci Sports Exerc* 36: 302–308, 2004.
20. Flanagan, T and Merrick, E. Holistic training: quantifying the work-load of soccer players. In: *Science and Football IV.* Spinks, W, Reilly, T, and Murphy, A, eds. London: Routledge, 2002. pp. 341–349.
21. Fox, EL and Mathews, DK. *Interval Training.* Paris: Vigot, 1977.
22. Gaiga, MC and Docherty, D. The effect of an aerobic interval training program on intermittent anaerobic performance. *Can J Appl Physiol* 20: 452–464, 1995.
23. Gaitanos, GC, Williams, C, Boobis, LH, and Brooks, S. Human muscle metabolism during intermittent maximal exercise. *J Appl Physiol* 75: 712–719, 1993.
24. Gupta, S, Goswami, A, Sadhukhan, AK, and Mathur, DN. Comparative study of lactate removal in short massage of extremities, active recovery and a passive recovery period after supramaximal exercise sessions. *Int J Sports Med* 17: 106–110, 1996.
25. Helgerud, J, Engen, LC, Wisløff, U, and Hoff, J. Aerobic endurance training improves soccer performance. *Med Sci Sports Exerc* 33: 1925–1931, 2001.
26. Hoff, J, Wisløff, U, Engen, LC, Kemi, OJ, and Helgerud, J. Soccer specific aerobic endurance training. *Br J Sports Med* 36: 218–221, 2002.
27. Impellizzeri, FM, Rampinini, E, and Marcora, SM. Physiological assessment of aerobic training in soccer. *J Sports Sci* 23: 583–592, 2005.
28. Karnoven, MJ, Kentala, E, and Mustala, O. The effect of training on heart rate, a longitudinal study. *Ann Med Exp Biol Fenn* 35: 307–315, 1957.
29. Kirkendall, DT. Training to mimic the game: in praise of 4 v 4. Master's thesis, University of North Carolina, Chapel Hill, 2001.
30. Léger, L and Boucher, R. An indirect continuous running multistage field test. The University of Montréal Track Test. *Can J Appl Sports Sci* 6: 77–84, 1980.
31. Mac Dougall, JD, Hicks, AL, MacDonald, JR, McKelvie, RS, Green, HJ, and Smith, KM. Muscle performance and enzymatic adaptations to sprint interval training. *J Appl Physiol* 84: 2138–2142, 1998.
32. Margaria, R, Olivia, RD, Di Prampero, PE, and Cerretelli, P. Energy utilisation in intermittent exercise of supramaximal intensity. *J Appl Physiol* 26: 752–756, 1969.
33. Millet, GP, Candau, R, Fattori, P, Bignet, F, and Varrav, A. VO_2 responses to different intermittent runs at velocity associated with VO_2max. *Can J Appl Physiol* 28: 410–423, 2003.
34. Nevill, ME, Williams, C, Ropper, D, Slater, C, and Nevill, AM. Effect of diet on performance during recovery from intermittent sprint exercise. *J Sports Sci.* 11: 119–126, 1993.

35. Rampinini, E, Impellizzerri, FM, Castagna, C, Abt, G, Chamari, K, Sassi, A, and Marcora, SM. Factors influencing physiological responses to small-sided games. *J Sports Sci* 25: 650–666, 2007.

36. Rodas, G, Ventura, JL, Cadefau, JA, Cusso, R, and Parra, J. A short training programme for the rapid improvement of both aerobic and anaerobic metabolism. *Eur J Appl Physiol* 82: 480–486, 2000.

37. Saltin, B and Essén, B. Muscle glycogen, lactate, ATP and PCr in intermittent exercise. In: Muscle Metabolism During Exercise: Advances in Experimental Medicine and Biology. Pernow, B and Saltin, B, eds. New York: Plenum Press, 1971. pp. 419–424.

38. Spalding, TW, Lyon, LA, Steel, DH, and Hatfield, BD. Aerobic exercise training and cardiovascular reactivity to psychological stress in sedentary young normotensive men and women. *Psychophysiology* 41: 552–562, 2004.

39. Stolen, T, Chamari, K, Castagna, C, and Wisløff, U. Physiology of soccer. An update. *Sports Med* 35: 501–536, 2005.

40. Thibault, G. Effects of moderate intensity endurance and high intensity intermittent training on anaerobic capacity and VO_2max. *Med Sci Sports Exerc* 28: 1327–1330, 1996.

41. Wisløff, U, Helgerud, J, and Hoff, J. Strength and endurance of elite soccer players. *Med Sci Sports Exerc* 30: 462–467, 1998.